Installation of Java and Execution of a Simple Hello World Program (Windows) in Oracle JDK

Software Required

- Windows Operating System
- Internet Connection
- Java Development Kit (JDK)

Step 1: Download Java JDK

1. Open a web browser.
2. Visit the official Oracle JDK website: Oracle JDK Downloads
3. Download the latest JDK for Windows (e.g., JDK 25).

Step 2: Install JDK

1. Double-click the downloaded .exe file.
2. Click **Next** and follow the installation wizard.
3. Complete the installation.

Step 3: Verify Java Installation

1. Open **Command Prompt**.
2. Type: java --version

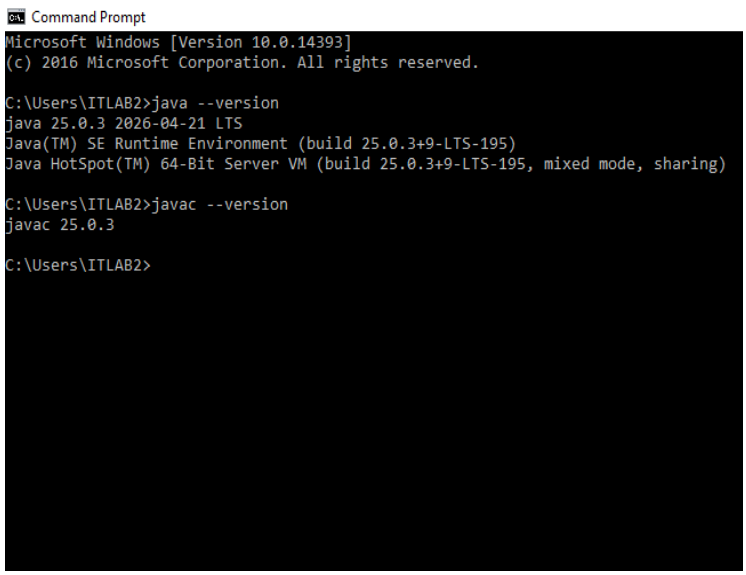
Sample Output:

java 25.0.3

3. Check the Java Compiler: javac --version

Sample Output:

javac 25.0.3



Configuring Java on Windows (Setting JAVA_HOME and PATH)

Step 1: Find Java Installation Path

After installing Java, locate the JDK folder.

Example:

C:\Program Files\Java\jdk-25.0.3

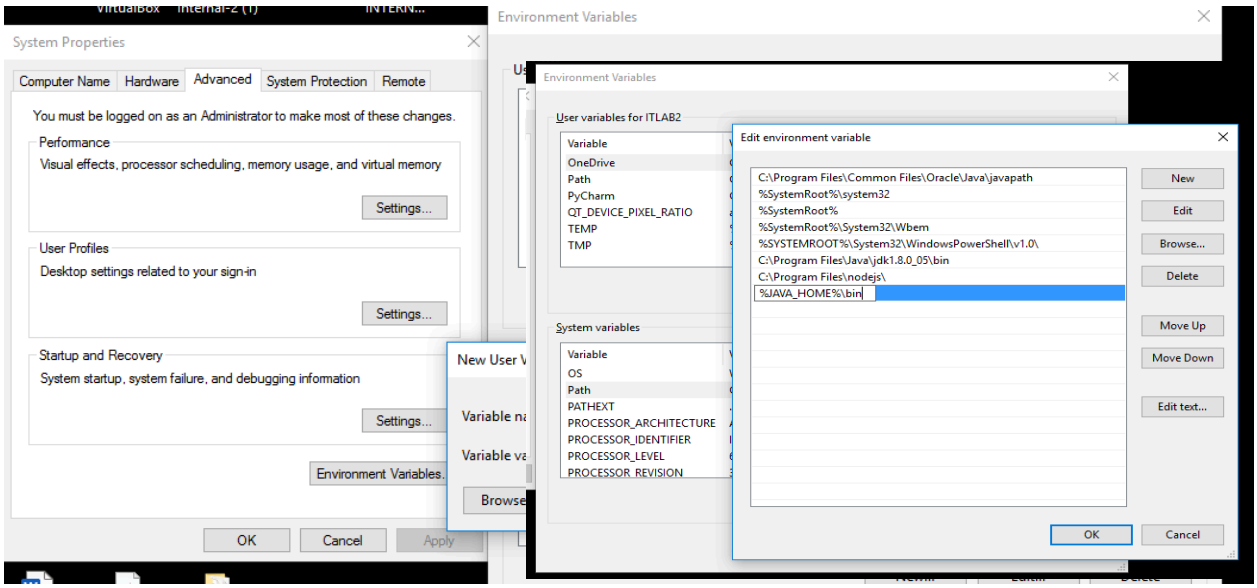
Step 2: Set JAVA_HOME

1. Press **Windows + R** → Type sysdm.cpl → Press Enter.
2. Select **Advanced** tab.

3. Click **Environment Variables**.
4. Under **System Variables**, click **New**.

Enter:

Variable Name : JAVA_HOME
Variable Value: C:\Program Files\Java\jdk-25.0.3



5. Click **OK**.

Step 3: Update PATH Variable

1. Under **System Variables**, select **Path**.
2. Click **Edit**.
3. Click **New**.
4. Add:

%JAVA_HOME%\bin

5. Click **OK → OK → OK**.

Step 4: Verify Configuration

Open a new Command Prompt and execute:

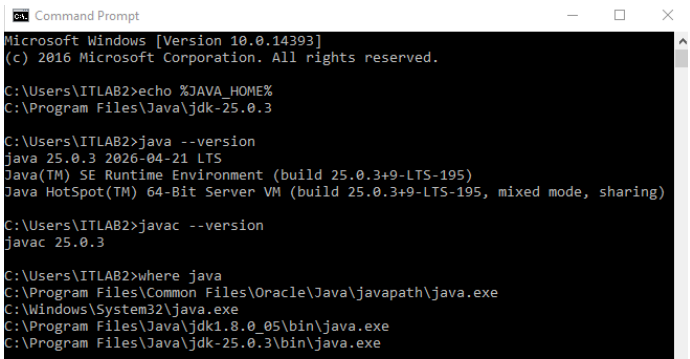
echo %JAVA_HOME%

Output:

C:\Program Files\Java\jdk-25.0.3

Check Java version:

java --version



Check Java compiler:

javac --version

Sample Output:

java 25.0.3
javac 25.0.3

Execution of ‘Hello’ program in Java:

1. Create a Java file:

Create a file named Hello.java:

```
public class Hello {
    public static void main(String[] args) {
        System.out.println("Hello");
    }
}
```

Store it in the format of .java

2. Open Command Prompt:

Type javac Hello.java

It would create a ‘.class’ file for the program.

3. Run the program:
Type java Hello
Sample Output:
Hello

